R pairs

qmd file can be right-click ==> open with text editor or quarto visual editor

R pairs

`pairs` R Scatterplot Matrix EDA

pairs

1. (i, j) i j
2.
   - `panel`
   - `diag.panel`
   - `upper.panel` / `lower.panel`
3. `iris`
4. `~ x + y + z`

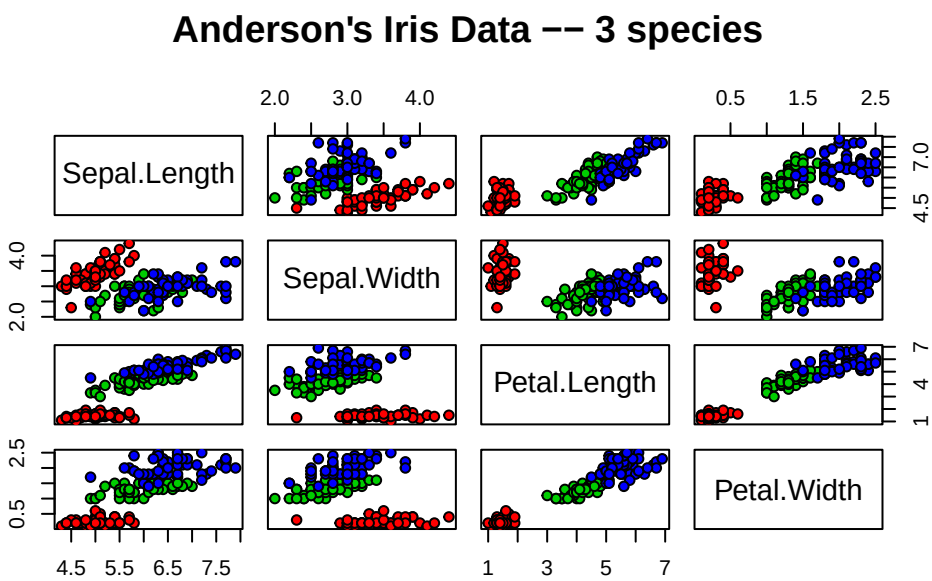
   `pairs`

1

iris

```
# iris[1:4]:   4
# main:
# pch = 21: plotting character   21          (col)   (bg)
# bg:
#     c("red", "green3", "blue")    palette
#     unclass(iris$Species)    1, 2, 3
```

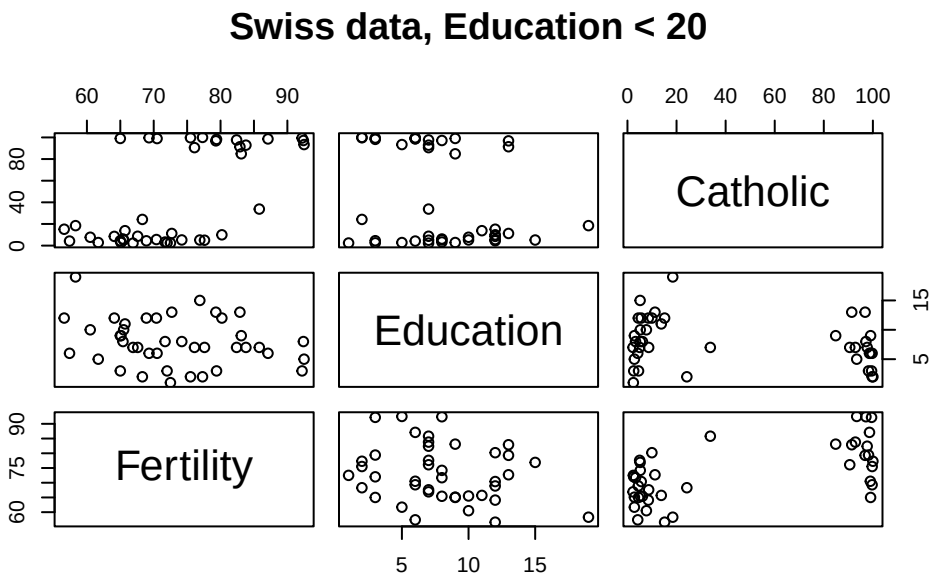
```r
pairs(
  iris[1:4],
  main = "Anderson's Iris Data -- 3 species",
  pch = 21,
  bg = c("red", "green3", "blue")[unclass(iris$Species)]
)
```



2

```r
#   : ~ Fertility + Education + Catholic
# data = swiss:
# row1attop = FALSE:
# subset = Education < 20:       Education   20
# main:
pairs(
  ~ Fertility + Education + Catholic,
  data = swiss,
  row1attop = FALSE,
  subset = Education < 20,
```

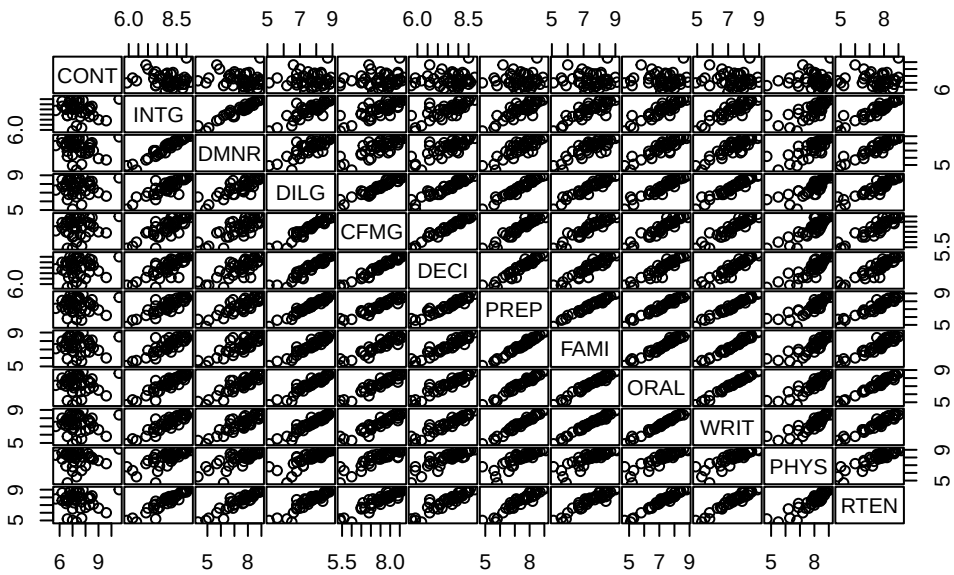
```
  main = "Swiss data, Education < 20"
)
```



3

gap

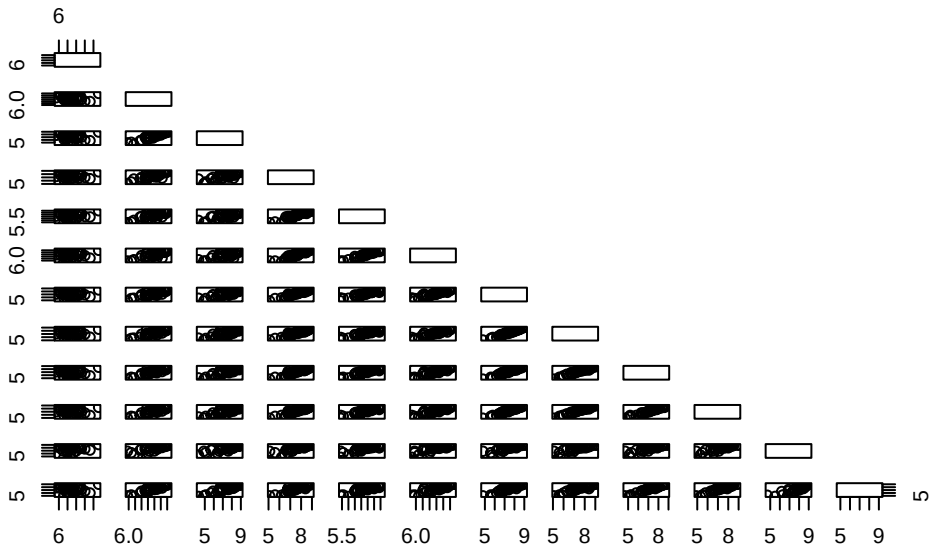
```
# USJudgeRatings: R
# gap = 1 / 10:
pairs(USJudgeRatings, gap = 1 / 10)
```



4

```
                upper.panel = NULL
```

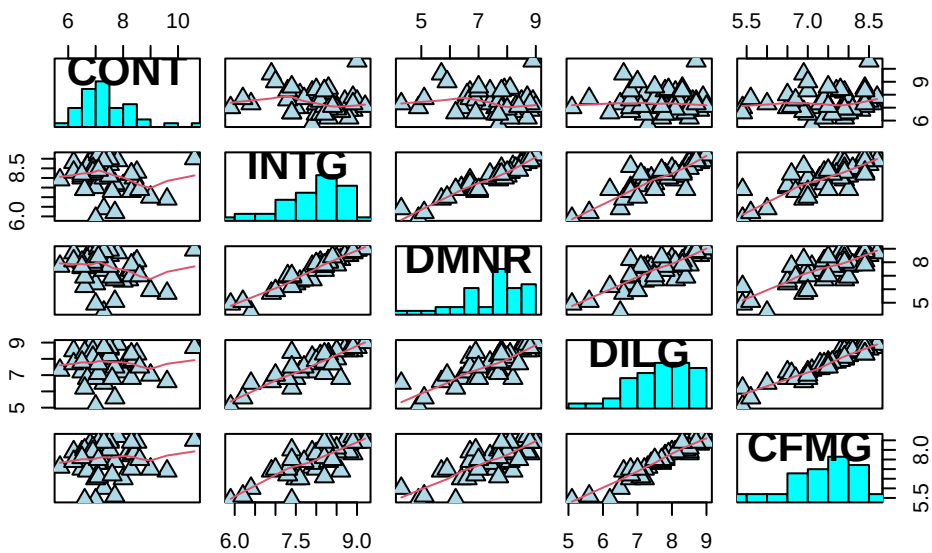
```r
# text.panel = NULL:
# upper.panel = NULL:
#
pairs(USJudgeRatings, text.panel = NULL, upper.panel = NULL)
```



5

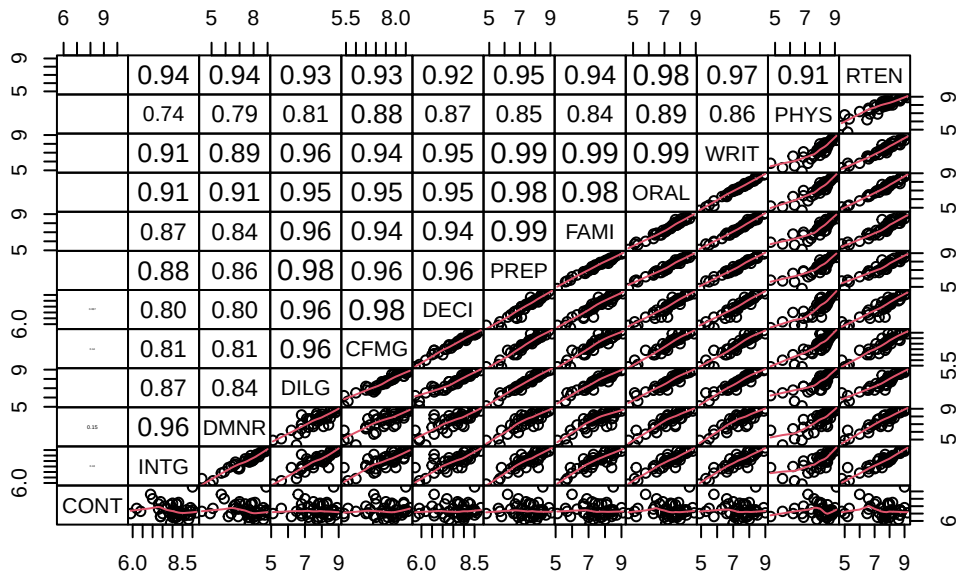
```
#
panel.hist <- function(x, ...) {
  usr <- par("usr") #
  #   y    0   1.5        x
  par(usr = c(usr[1:2], 0, 1.5))
  h <- hist(x, plot = FALSE) #
  breaks <- h$breaks
  nB <- length(breaks)
  y <- h$counts
  y <- y / max(y) #      [0, 1]
  #
  rect(breaks[-nB], 0, breaks[-1], y, col = "cyan", ...)
}

#      pairs
# USJudgeRatings[1:5]:   5
# panel = panel.smooth:       loess
# cex = 1.5:
```

```r
# pch = 24:
# bg = "light blue":
# horOdd = TRUE:
# diag.panel = panel.hist:
# cex.labels = 2:
# font.labels = 2:       2
pairs(
  USJudgeRatings[1:5],
  panel = panel.smooth,
  cex = 1.5,
  pch = 24,
  bg = "light blue",
  horOdd = TRUE,
  diag.panel = panel.hist,
  cex.labels = 2,
  font.labels = 2
)
```



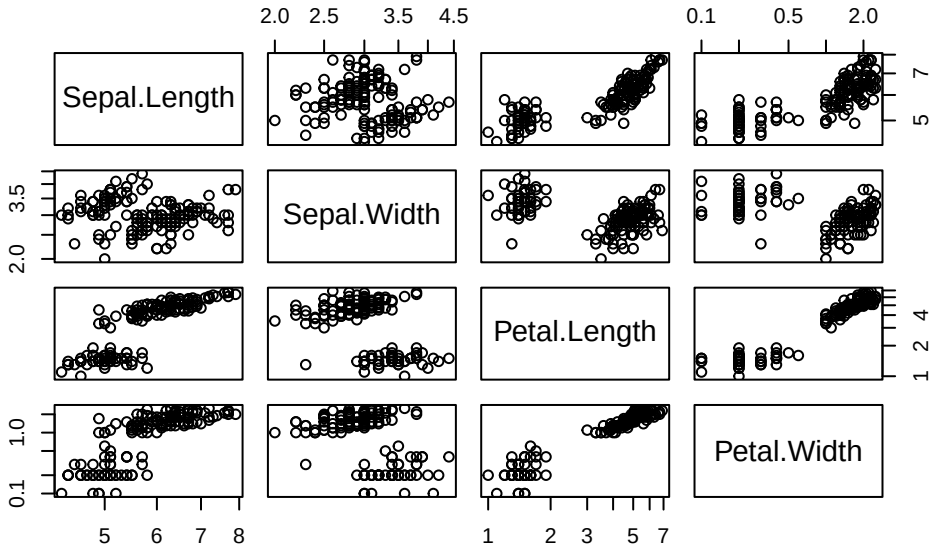
6

```r
#
panel.cor <- function(x, y, digits = 2, prefix = "", cex.cor, ...) {
  par(usr = c(0, 1, 0, 1)) #      [0,1] x [0,1]
  r <- abs(cor(x, y)) #
  #
  txt <- format(c(r, 0.123456789), digits = digits)[1]
  txt <- paste0(prefix, txt)

  #     cex.cor
  #
  if (missing(cex.cor)) {
    cex.cor <- 0.8 / strwidth(txt)
  }
  #
  # cex = cex.cor * r
  text(0.5, 0.5, txt, cex = cex.cor * r)
}

#          pairs
# lower.panel = panel.smooth:
# upper.panel = panel.cor:
# gap = 0:
# row1attop = FALSE:
pairs(
  USJudgeRatings,
  lower.panel = panel.smooth,
  upper.panel = panel.cor,
  gap = 0,
  row1attop = FALSE
)
```

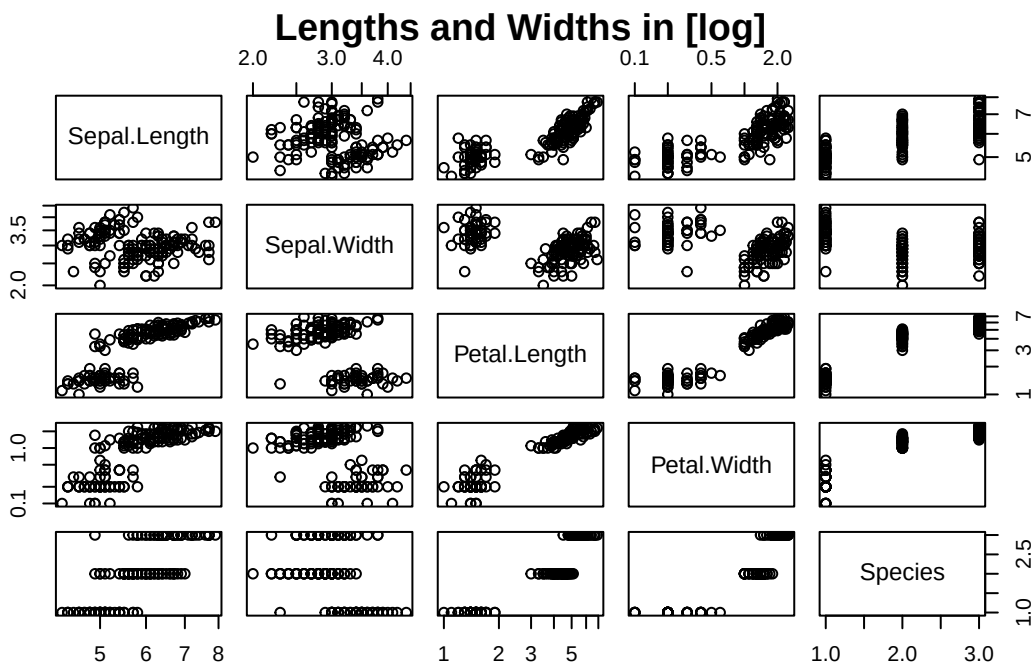


7

```
# log = "xy":     x   y
# iris[-5]:   5 Species
pairs(iris[-5], log = "xy")
```




```
# log = 1:4:   4   1 4
# main:
# line.main = 1.5:
# oma = c(2, 2, 3, 2):
pairs(
  iris,
  log = 1:4, #
  main = "Lengths and Widths in [log]",
  line.main = 1.5,
  oma = c(2, 2, 3, 2)
)
```


Lengths and Widths in [log]
Sepal.Length
Sepal.Width
Petal.Length
Petal.Width
Species
2.0
3.0
4.0
0.1
0.5
2.0
7
5
3.5
2.0
7
3
1
1.0
0.1
2.5
1.0
5
6
7
8
1
2
3
5
1.0
2.0
3.0

`pairs` R